

TIANLE WANG

✉ wtl233wtl@gmail.com · ☎ (+1) 858-531-8206 · 🌐 wtl666wtl · 🏠 Home Page

EDUCATION

Purdue University

West Lafayette, IN, USA

Ph.D. in Computer Science

Aug. 2025 – Present

- Advised by Prof. Abulhair Saparov
- GPA: 4.0/4.0

University of California, San Diego (UCSD)

La Jolla, CA, USA

Master of Science in Data Science

Sept. 2023 – Mar. 2025

- Advised by Prof. Jingbo Shang
- GPA: 4.0/4.0

Shanghai Jiao Tong University (SJTU)

Shanghai, China

Bachelor of Engineering in Computer Science

Sept. 2019 – Jun. 2023

- Member of **ACM Honors Class**, which is an elite program for top 5% talented students
- GPA: 3.7/4.3

PUBLICATION

1. **Tianle Wang**, Zihan Wang, Weitang Liu and Jingbo Shang “WOT-Class: Weakly Supervised Open-world Text Classification”, *CIKM 2023*
2. Zihan Wang*, **Tianle Wang***, Dheeraj Mekala and Jingbo Shang “A Benchmark on Extremely Weakly Supervised Text Classification: Reconcile Seed Matching and Prompting Approaches”, *ACL 2023, Findings*
3. Yiyang Zhou*, Zhaoyang Wang*, **Tianle Wang***, Shangyu Xing, Peng Xia, Bo Li, Kaiyuan Zheng, Zijian Zhang, Zhaorun Chen, Wenhao Zheng, Xuchao Zhang, Chetan Bansal, Weitong Zhang, Ying Wei, Mohit Bansal, Huaxiu Yao, “AnyPrefer: An Automatic Framework for Preference Data Synthesis”, *ICLR 2025*
4. Tianxing He, Jingyu Zhang, **Tianle Wang**, Sachin Kumar, Kyunghyun Cho, James Glass and Yulia Tsvetkov “On the Blind Spots of Model-Based Evaluation Metrics for Text Generation”, *ACL 2023*
5. Mengke Zhang, Tianxing He, **Tianle Wang**, Lu Mi, Fatemehsadat Mireshghallah, Binyi Chen, Hao Wang, Yulia Tsvetkov “LatticeGen: A Cooperative Framework Which Hides Generated Text in A Lattice For Privacy-Aware Generation on Cloud”, *NAACL 2024, Findings*
6. Guangchen Lan, Sipeng Zhang, **Tianle Wang**, Yuwei Zhang, Daoan Zhang, Xinpeng Wei, Xiaoman Pan, Hongming Zhang, Dong-Jun Han, Christopher G Brinton “MaPPO: Maximum a Posteriori Preference Optimization with Prior Knowledge”, *arXiv 2025, under review*
7. **Tianle Wang**, Zhaoyang Wang, Guangchen Lan, Xinpeng Wei, Sipeng Zhang, GuanWen Qiu, Abulhair Saparov “Can RL Teach Long-Horizon Reasoning to LLMs? Expressiveness Is Key”, *under review*

RESEARCH EXPERIENCE

University of North Carolina at Chapel Hill

Chapel Hill, NC, USA

Research Intern, advised by **Prof. Huaxiu Yao**

Jun. 2024 – Dec. 2024

- Self-evolving framework for preference data generation
 - Introduced a novel framework Anyprefer, which synthesizes high-quality preference data for foundation models, improving alignment with human values across diverse modalities
 - Implemented a cooperative Markov Game-based feedback mechanism to jointly enhance the quality of generated preference data and improve reward estimation accuracy
 - Accepted by *ICLR 2025* as co-first author

SLS Lab, Massachusetts Institute of Technology

Research Intern, advised by **Prof. James Glass** and **Dr. Tianxing He**

Cambridge, MA, USA

Jul. 2022 – Jan. 2023

- Blindsports in PLM-based NLG Metrics
 - Revealed the lack of robustness in PLM-based NLG metrics (e.g., BERTScore) and created an evaluation system with synthetic data for comprehensive checks
 - Identified biases and flaws in various metrics, proposing solutions to mitigate undesirable behaviors
 - Accepted by *ACL 2023*

SDLab, University of California, San Diego

Research Intern, advised by **Prof. Jingbo Shang**

La Jolla, CA, USA

Aug. 2021 – Dec. 2024

- Weakly Supervised Open-world Text Classification
 - Pioneered weakly supervised open-world text classification, revealing limitations of existing methods
 - Proposed WOT-Class, which jointly improves class identification and document classification, achieving an over 20% macro-F1 improvement across datasets
 - Accepted by *CIKM 2023* as first author
- Benchmark for Extremely Weakly Supervised Text Classification
 - Developed a benchmark to compare extremely weakly supervised text classification methods, particularly between representation learning and prompt-based approaches
 - Conducted a thorough analysis of key factors influencing model performance, including dataset characteristics, prompt design strategies, class name and language model selection
 - Accepted by *ACL 2023* as co-first author

SELECTED HONORS AND AWARDS

Irving T. Ho Memorial Scholarship

One of the 4 winners in SJTU

2022

Zhiyuan Honorary Scholarship

Top 5% in SJTU

2019 - 2022

Silver Medal

Top 150 in The 34th China's National Olympiad in Informatics (NOI)

2017

TEACHING EXPERIENCE

Instructional Assistant

Fall 2024

Advanced Data-Driven Text Mining (CSE 291 & DSC 253), UCSD

Teaching Assistant

Spring 2021

Data Structures (CS 152), SJTU

Teaching Assistant

Fall 2020

Great Ideas in Computer Science (CS 163), SJTU

SELECTED PROJECTS

🔗 x-TC

Fall 2023

Benchmark, Text Classification

- An implementation of our *ACL* paper
- Built a benchmark for comparing different extremely weakly supervised text classification methods
- User-friendly to edit the framework and incorporate novel methods and datasets

🔗 WOT-Class

Spring 2023

Text Classification, Open-world Learning

- An implementation of our *CIKM* paper
- Designed an iterative refinement framework that achieved state-of-the-art performance in weakly supervised open-world text classification settings